SET tidal response

Burden gradients and the relative motion of nearby probes

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Two nearby objects can fall differently while occupying almost the same field. The difference depends on how the gravitational response changes between them. SET describes that response through the spatial variation of burden. Here we follow the connection from the capacity relation to acceleration, relative motion and tides.

$$S^2 + |\mathbf{V}|^2 + c^2\beta^2 = c^2, \qquad \beta = \frac{d\tau}{dT} > 0. \tag{1}$$

Gravitational burden S is expressed in speed units. Translation $\mathbf{V}$ and event progression β share the remaining capacity. For a held clock, define α; for the gravitational energy reading, define U:

$$\alpha^2 = 1 - \frac{S^2}{c^2}, \qquad U = \frac{S^2}{2}, \qquad L_p = -mc^2\beta. \tag{2}$$

The last expression is SET's retained probe coupling, with calibrated rest mass m. Use a stationary field and nonrotating Cartesian spatial readings at weak-field accuracy, so $\mathbf{V} = \mathrm{d}\mathbf{x}/\mathrm{d}T$. Expanding for $S^2/c^2 \ll 1$ and $|\mathbf{V}|^2/c^2 \ll 1$ gives

$$L_p = -mc^2 + \frac{m}{2}|\mathbf{V}|^2 + mU + O\left(\frac{m(S^2 + |\mathbf{V}|^2)^2}{c^2}\right). \tag{3}$$

The Euler–Lagrange equation of this leading action is $m\mathrm{d}\mathbf{V}/\mathrm{d}T = m\nabla U$. Thus, to the retained weak-field order,

$$\mathbf{a} = \nabla U = \frac{1}{2}\nabla S^2 = -\mathbf{J}, \qquad \mathbf{J} \equiv -\nabla U. \tag{4}$$

Let the two freely moving probes lie at $\mathbf{x}$ and $\mathbf{x} + \boldsymbol{\xi}$. Taylor expansion of their acceleration difference gives

$$\delta a_i = a_i(\mathbf{x} + \boldsymbol{\xi}) - a_i(\mathbf{x}) = \mathcal{T}_{ij}\xi_j + O(|\boldsymbol{\xi}|^2), \tag{5}$$

$$\mathcal{T}_{ij} \equiv \partial_i\partial_j U = \frac{1}{2}\partial_i\partial_j S^2 = -\partial_j J_i = -\frac{c^2}{2}\partial_i\partial_j\alpha^2. \tag{6}$$

Repeated spatial indices are summed. The symmetric tensor $\mathbf{T}$ has units s^{-2}: multiplied by a separation, it gives a relative acceleration. Its algebraic clock and burden readings are identical; interpreting it as the acceleration gradient uses the probe coupling above. The separation obeys $\mathrm{d}^2\boldsymbol{\xi}/\mathrm{d}T^2 = \mathbf{T}(\mathbf{x}(T))\boldsymbol{\xi}$ to first order.

The source law remains visible in the spherical calculation. For a spherical source domain with radius R_0 and reference volume V_Ω, write

$$Q_\Omega^2 = 24\pi G M V_\Omega, \qquad V_\Omega = \frac{4\pi R_0^3}{3}, \qquad \mu \equiv GM = \frac{Q_\Omega^2}{32\pi^2 R_0^3}. \tag{7}$$

Use the retained isolated spherical exterior, with the unloaded reference at infinity. At a measurement radius r outside the source,

$$U(r) = \frac{\mu}{r}, \qquad S^2(r) = \frac{2\mu}{r}, \qquad Q_\Omega = 4\pi R_0^2 S(R_0). \tag{8}$$

Here μ is fixed by the source law, while $U = \mu/r$ is the retained exterior response. The fixed Q_Ω belongs to R_0; $4\pi r^2 S(r)$ is a different, radius-dependent screen reading.

With $r = |\mathbf{x}|$ and $\mathbf{n} = \mathbf{x}/r$, direct differentiation gives

$$\partial_i U = -\frac{\mu x_i}{r^3}, \qquad \partial_j \partial_i U = \mu\left(\frac{3x_i x_j}{r^5} - \frac{\delta_{ij}}{r^3}\right). \tag{9}$$

$$\mathcal{T}_{ij} = \frac{Q_\Omega^2}{32\pi^2 R_0^3 r^3}(3n_i n_j - \delta_{ij}). \tag{10}$$

The source radius R_0 and measurement radius r remain distinct. The radial and transverse eigenvalues are

$$\mathcal{T}_{\text{radial}} = \frac{2\mu}{r^3}, \qquad \mathcal{T}_{\text{transverse}} = -\frac{\mu}{r^3}, \qquad \operatorname{tr}\mathcal{T} = 0. \tag{11}$$

Radially separated probes stretch apart because the inner probe accelerates inward more strongly. Transversely separated probes converge because their inward acceleration vectors point toward the same center. A radial derivative alone would miss that second effect.

The three coefficients sum to zero in this source-free inverse-radius exterior, although the burden remains nonzero.

The finite current gives the same result before taking a derivative. Along a stated straight spatial coordinate χ, let $J_- = (U_- - U_0)/L$ and $J_+ = (U_0 - U_+)/L$, using equally spaced points. Then

$$\mathcal{T}_{\chi\chi}^{(L)} \equiv -\frac{J_+ - J_-}{L} = \frac{S_+^2 - 2S_0^2 + S_-^2}{2L^2} \longrightarrow \mathcal{T}_{\chi\chi}. \tag{12}$$

The finite expression is exact as a defined reading. Its limit is the directional second derivative; for a smooth field the centered approximation has an error of order L^2. A change in the sampling distance must carry its corresponding derivative conversion.

Cancellation of the pull does not cancel the response to displacement. Consider two equal spherical sources at $\pm d\mathbf{e}_x$, each with gravitational coefficient μ. Use the additive isotropic weak-field source model already used in the Lagrange calculation:

$$U(\mathbf{x}) = \frac{\mu}{|\mathbf{x} - d\mathbf{e}_x|} + \frac{\mu}{|\mathbf{x} + d\mathbf{e}_x|}. \tag{13}$$

This addition rule specifies the two-source comparison. At the midpoint, symmetry gives

$$S^2(0) = \frac{4\mu}{d}, \qquad \nabla U(0) = 0, \qquad \alpha_0^2 = 1 - \frac{4\mu}{dc^2}. \tag{14}$$

For $\mu/(dc^2) \ll 1$, both the field and the motion treatment are weak field. The two acceleration vectors cancel exactly at the midpoint, while their derivatives add:

$$\mathcal{T}(0) = \frac{\mu}{d^3}\mathrm{diag}(4, -2, -2). \tag{15}$$

A small displacement along the source axis therefore grows away from the midpoint if the sources are held fixed. A transverse displacement is pulled back toward that axis. The midpoint has a finite burden, zero acceleration and a nonzero tidal response, all within the same specified field.

This is the precise sense in which the field remains occupied where directional effects balance. It does not require interpreting the canceled vectors as canceled amounts of burden. Nor does it establish that burden is transported to another boundary; that would concern the evolution of the sources and field.

If the sources orbit, the comparison must include rotation. In coordinates rotating at constant $\mathbf{\Omega}$, define the effective potential reading $F = U + |\mathbf{\Omega} \times \mathbf{r}|^2/2$. Linearizing the retained weak-field motion about an equilibrium gives

$$\delta\ddot{\mathbf{r}} + 2\mathbf{\Omega} \times \delta\dot{\mathbf{r}} = H_F\, \delta\mathbf{r}, \qquad H_F = \mathcal{T} + \Omega^2 I - \mathbf{\Omega}\mathbf{\Omega}^T. \tag{16}$$

Dots refer to T. The Coriolis term acts on velocity, so the eigenvalues of the gravitational tidal tensor alone cannot decide the stability of a rotating equilibrium. The fuller capacity expression also retains the inertial response derived in *SET inertia*; equation (16) is its weak-field limit.

For the equal pair on circular orbits, $\Omega^2 = \mu/(4d^3)$. At the midpoint, with the rotation axis along z, $\mathbf{H}_F/\Omega^2 = \mathrm{diag}(17, -7, -8)$. A planar displacement proportional to $\exp(\lambda\Omega T)$ consequently obeys

$$\lambda^4 - 6\lambda^2 - 119 = 0, \qquad \lambda^2 = 3 \pm 8\sqrt{2}. \tag{17}$$

One pair of modes grows and decays exponentially. This is the midpoint Lagrange instability in the stated circular model. It follows after retaining the rotational response, rather than treating a zero gradient as sufficient for stability.

Orbital motion reads both the gradient and how that gradient changes. In a stationary axisymmetric weak field, take the equatorial plane to be a reflection symmetry plane. A circular orbit at cylindrical radius R satisfies

$$v_c^2 = -R\,\partial_R U, \qquad \Omega_c^2 = -\frac{\partial_R U}{R}. \tag{18}$$

For a small radial perturbation, conserve the specific angular momentum j. The radial equation is $\ddot{R} = \partial_R U + j^2/R^3$. Differentiating its right-hand side at the circular orbit, where $j^2 = \Omega_c{}^2 R^4$, yields

$$\delta\ddot{R} = \left(\partial_R^2 U - 3\Omega_c^2\right)\delta R, \qquad \kappa^2 = 3\Omega_c^2 - \mathcal{T}_{RR}. \tag{19}$$

Here κ is the radial oscillation frequency when $\kappa^2 > 0$. The equatorial vertical frequency is similarly $\nu_z{}^2 = -\partial_z{}^2 U$. In the spherical exterior, $\mathbf{T}_{RR} = 2\Omega_c{}^2$, so $\kappa = \Omega_c$. For a field with constant circular speed over an interval, $\partial_R U = -\nu_0{}^2/R$, and $\kappa = \sqrt{2}\,\Omega_c$. These are weak-field profile diagnostics; the second case stipulates a flat-speed interval rather than deriving it from a galaxy's sources.

Thus, matching a circular speed at one radius fixes a first derivative there. Relative motion and orbital oscillations provide a way to examine the next derivative. They contain information that a single speed does not.

Clock readings provide another expression of the same burden variation. For three held clocks at the equally spaced points used in equation (12), all normalized to the same reference time, substitution of $S^2 = c^2(1 - \alpha^2)$ gives

$$\mathcal{T}^{(L)}_{\chi\chi} = -\frac{c^2}{2L^2}\left(\alpha_+^2 - 2\alpha_0^2 + \alpha_-^2\right). \tag{20}$$

This is an exact conversion of the finite reading. Converting clock comparisons into a numerical tensor also requires the distance calibration and common clock normalization. Clock ratios alone do not fix an absolute external time scale.

The connection with light uses the stationary operational coupling already retained in SET:

$$\frac{d\ell}{dT} = c\alpha, \qquad \frac{d\tau_h}{dT} = \alpha, \qquad \frac{d\ell}{d\tau_h} = c. \tag{21}$$

The clock participates in the field it measures. A held observer reads local light speed c, while comparisons between locations can read the change in α. The local distance ℓ in equation (21) must be converted to the spatial sampling coordinate before taking a derivative. At the weak-field order used above, the nonrotating Cartesian reading supplies that conversion.

This gives a consistent role to our optical work: signals can compare clock progression, while trajectories compare gradients. No additional electromagnetic response is needed for the identities in this section, and no new electromagnetic evolution equation is being selected by them.

The black-hole comparison requires the retained relativistic spatial response. The Cartesian Hessian above is a weak-field acceleration gradient. To discuss a horizon, use the spherical interval already retained in the perihelion calculation:

$$c^2 d\tau^2 = c^2 f(r) dT^2 - \frac{dr^2}{f(r)} - r^2 d\Omega^2, \qquad f = \alpha^2 = 1 - \frac{S^2}{c^2} = 1 - \frac{2\mu}{c^2 r}. \tag{22}$$

Now relative accelerations are measured in an orthonormal frame carried by a radial freely falling observer, per that observer's proper time. Evaluating the tidal response of this interval gives the radial and two equal transverse coefficients [4]:

$$\Lambda_r = -\frac{c^2}{2} f'' = \frac{1}{2}(S^2)'' = \frac{2\mu}{r^3}, \qquad \Lambda_t = -\frac{c^2}{2r} f' = \frac{(S^2)'}{2r} = -\frac{\mu}{r^3}. \tag{23}$$

Primes denote derivatives with respect to the areal radius. For this interval, the radial boost from a local static frame to a radial free-fall frame leaves these coefficients unchanged: the transverse temporal and radial curvature terms are equal and opposite. The calculation uses the full retained interval; it is not an extrapolation of coordinate acceleration to the horizon.

Let $R_s = 2\mu/c^2$. The exterior field has $S(R_s) = c$, while the limiting tidal coefficients are

$$\Lambda_r(R_s) = \frac{c^2}{R_s^2}, \qquad \Lambda_t(R_s) = -\frac{c^2}{2R_s^2}. \tag{24}$$

Two horizons can therefore have the same limiting burden and different tides. For illustrative nonrotating masses of 10 and 10^6 solar masses, the horizon radii are 29.53 km and 2.953×10^6 km. Their radial coefficients are 1.030×10^8 s^{-2} and 1.030×10^{-2} s^{-2}, respectively [1]. These are calculated free-fall limits, not measurements by a clock held at a horizon; no such static clock exists there.

The same distinction clarifies the Kretschmann scalar. In this spherical vacuum branch, the curvature calculation gives

$$K = \frac{48\mu^2}{c^4 r^6} = \frac{8}{c^4}\left(\Lambda_r^2 + 2\Lambda_t^2\right) = \frac{12 S^4}{c^4 r^4}. \tag{25}$$

Since each tidal coefficient has units s^{-2}, dividing its squared norm by c^4 gives m^{-4}. The units describe a squared inverse-area curvature scale. In SET variables, K is a geometric reading of the spatial response associated with burden; it is not determined by the burden value alone.

$$K = \frac{3 Q_\Omega^4}{64\pi^4 R_0^6 c^4 r^6}, \qquad K(R_s) = \frac{12}{R_s^4}. \tag{26}$$

The source and measurement radii again remain separate. The finite horizon value follows from the retained spherical geometry. It neither defines an interior continuation of the static capacity reading nor removes the central singularity of that geometry.

A terrestrial gradient gives the calculation a measured scale. Represent Earth by the spherical reference radius $R_0 = 6.371 \times 10^6$ m and use the DE440 gravitational parameter $\mu = 3.98600435507 \times 10^{14}$ m³ s⁻² [1, 2]. The source and surface burden readings are

$$Q_\Omega = 4\pi\sqrt{2\mu R_0^3} = 5.706 \times 10^{18}\ \mathrm{m^3\,s^{-1}}, \qquad S(R_0) = 11.186\ \mathrm{km\,s^{-1}}. \tag{27}$$

The predicted radial tidal coefficient is $2\mu/R_0^3 = 3.0828 \times 10^{-6}$ s⁻². For two vertically separated probes, the upper one falls downward less strongly. If g denotes the positive downward magnitude and z increases upward, the reported gradient is $\mathrm{d}g/\mathrm{d}z = -\mathbf{T}_{zz}$. Keeping that sign convention,

Quantity	Value
Spherical SET reading dg/dz	-3.083×10^{-6} s⁻²
NIST laboratory determination [3]	Approximately -3×10^{-6} s⁻²
Spherical radial separation response for a 1 m baseline	$+3.083 \times 10^{-6}$ m s⁻²

The finite one-metre result can be checked without differentiating:

$$\delta a_z = \mu\left[\frac{1}{R_0^2} - \frac{1}{(R_0+L)^2}\right] = \frac{\mu L(2R_0+L)}{R_0^2(R_0+L)^2}, \qquad L = 1\ \mathrm{m}. \tag{28}$$

It gives 3.082796×10^{-6} m s⁻². Bartel's NIST report describes the gravity-gradient determination from free-fall measurements and quotes the rounded value in the table, without a separate uncertainty for that gradient in the cited passage. The sphere is not a reconstruction of the laboratory's local terrain, rotation, nearby structures or Earth's multipoles. The comparison establishes the observed sign and scale; it does not establish precision agreement or distinguish SET from the Newtonian limit it retains.

The corresponding centered one-metre clock reading in the spherical model is $\alpha_+^2 - 2\alpha_0^2 + \alpha_-^2 = -6.860 \times 10^{-23}$. This is a calculated implication of equation (20), not a reported clock measurement.

SET separates three readings of the same field. Burden enters the capacity available for local events. Its gradient governs weak-field probe acceleration. The variation of that gradient determines how nearby motions separate or converge. A balanced pull can coexist with burden and tides; equal burden values can coexist with different tidal strengths. These distinctions follow through the retained probe and spherical response without introducing a separate tidal interaction.

[1] JPL Solar System Dynamics, Astrodynamic Parameters, DE440 values. Solar GM = 1.32712440041279419 × 10^{20} m³ s⁻²; c = 299,792,458 m s⁻¹. [2] NASA NSSDCA, Earth Fact Sheet, volumetric mean radius. Accessed 12 October 2026.

[3] T. Bartel, Uncertainty in NIST Force Measurements, J. Res. NIST 110, 589–603 (2005), section 3.2, p. 593. [4] Tidal forces in dirty black hole spacetimes, Eur. Phys. J. C (2022), section 2.3 and its Schwarzschild limit.

Verification: 32 symbolic identities and 900 numerical derivative components checked. The numerical Hessian checks had maximum error 2.93×10^{-9} relative to the local μ/r^3 scale. These checks validate the calculation, separately from the observational comparison.